

Architectural Particular Specification

DAMPPROOFING AND WATERPROOFING

SECTION 6.1

Rev	Date	Description
0	14 APR 2022	First Issue
A	13 MAY 2022	Tender Addendum No.2
B	05 JUL 2022	Post Tender Addendum No.1

SECTION 6.1

DAMPPROOFING AND WATERPROOFING

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The Work of this Section includes, but is not limited to, the furnishing and installation of the following :
1. Waterproofing membrane and coatings and required ancillary materials at external and internal areas indicated on the Drawings and Specification.
 2. Protection boards, water stops and other required accessories.
 3. Tank linings and required accessories.
 4. Leak, compatibility, material and other testings.
 5. Service entry penetration waterproofing through tankings and water retaining structures.
 6. The Management Contractor shall provide equipment, plant, water and drainage required for all testing of water retaining structures, tanks, and waterproofing work.
 7. Coordination and interface with all local utility company requirements, services and penetrations through all tanks and water retaining structures. Refer to Consulting Engineers' Contract Documents.
 8. Site photography of waterproofing and penetration sealing Work.
- B. Related Sections :
1. Section 2.1: CAST DECKS AND UNDERLAYMENT
 2. Section 6.2: THERMAL PROTECTION INSULATION
 3. Section 6.5: JOINT SEALANTS
 4. Section 8.3: PAINTING
 5. Section 8.4: TILING
 6. Section 8.9: SPECIAL COATINGS
 7. Consulting Engineers' Contract Documents for site drainage, tanks and water retaining structures, plumbing, drainage and sewage systems, and other related works.

1.02 REFERENCES

- A. British Standards :
1. BS 6398 : 1989 - Specification for Bitumen Damp-Proof Courses for Masonry.
 2. BS 6068 : All Applicable Parts and Sections - Water Quality : Physical, Chemical, Microbiological, Biological And Biochemical Test Methods And Sampling.
 3. BS 6920 : Suitability Of Non-Metallic Products for Use in Contact With Water Intended For Human Consumption With Regard to Their Effect on The Quality of Water.
 4. BS 8000 : Part 4 : 1989 - Code of Practice for Waterproofing

5. BS 8007 : 1987 - Code of Practice For Design of Concrete Structures for Retaining Aqueous Liquids.
6. BS 8102: 1990 - Code of Practice for Protection of Structures Against Water From The Ground.
7. BS 8215: 1991 - Code of Practice for Design And Installation of Damp-Proof Courses in Masonry Construction.
Part 1: 1990 – Specification
Part 2 : 1990 - Methods of Test.
8. Construction Product List A, Part 2, 1.10 by DIBt – Notes for the application of waterproofing directly under tiles for indoor and outdoor use.

B. American Standards

1. ASTM C898-89 : Guide for Use of High Solids Content, Cold Liquid - Applies Elastomeric Waterproof Membrane With Separate Wearing Course.
2. ASTM Publication STP 1084 - Building Deck Waterproofing, 1990.
3. American Concrete Institute ACI Publication 515. IR - Guide to The Use of Waterproofing, Damp proofing, Protective and Decorative Barrier Systems for Concrete, 1985.
4. National Roofing Contractors' Association - NRCA Roofing And Waterproofing Manual, 4th Edition, 1996.
5. ASTM, C-981 - Standard Guide for Design of Built-up Bituminous Membrane Waterproofing Systems for Building Decks .
6. ASTM C836 - Standard Specification for High Solids Content, Cold Liquid-Applied Elastomeric Waterproofing Membrane for Use with Separate Wearing Course.
7. ASTM D412 - Standard Test Methods for Vulcanized Rubber and Thermoplastic Rubbers and Thermoplastic Elastomers. Tensile strength and elongation tests.
8. ASTM E96 - Standard Test Method for Water Vapour Transmission of Materials. Method BW.
9. ASTM D2240 - shore A hardness testing
10. ASTM G 26 – weather-meter testing

1.03 QUALITY ASSURANCE

A. Qualification of Waterproofing specialist sub-contractor :

Company/ies specializing in the installation of the waterproofing and/or tank lining systems specified herein, with minimum of ten (10) years documented experience (most recent five (5) years prior to tender date), certified and approved by the waterproofing manufacturers specified herein.

B. Qualification of Waterproofing Manufacturers.

1. Specializing in the manufacturing of high-performance, commercial grade waterproofing system and materials for at least ten (10) years.
2. The Waterproofing Material Manufacturers shall employ a full-time Field Technical Representatives available for interim and final Project inspections. The Manufacturer's Field Technical Representative shall be experienced and knowledgeable in the waterproofing material and its application.
3. The Waterproofing Manufacturer's shall be capable of providing Field Service Representation during construction, approving acceptable installer and application method, and conducting final inspection of the waterproofing system/assembly.

4. Submit Manufacturer's compatibility testing, field reports on substrate, application and final inspections of installed waterproofing.
- C. Project Record Documentation
1. The Contractor shall photograph installation of waterproofing, tank lining, and penetrating sealing work as part of the Project Record Documentation.

1.04 PERFORMANCE REQUIREMENTS

- A. All waterproofing and tank lining waterproofing systems, material, design and installation work shall strictly conform to BS 8000 : Part 4 and BS 8102 : 1990. Tank lining for retaining potable water shall conform to BS 6920 : 1990 (WRC approved) and be approved for use in Hong Kong by the appropriate authorities. Lining sewage filtration and holding tanks shall be resistant to aerobic and anaerobic bacteriological attacks and chemicals used to disinfect tanks.
- B. All waterproofing and tank lining shall only be carried out after the concrete has been properly cured for at least 7 days.
- C. Clean and sterilize tanks and water retaining structures as required and specified in this Section.
- D. Watertightness of structures and water sterilization testing, sampling and monitoring procedures shall conform to this Specification and in accordance with BS 6068 and BS 6920 Part 1 and Part 2.
- E. Provide and install membrane roofing system materials that have been tested and listed by Underwriter's Laboratory (UL) in UL Building Materials Directory for applications indicated, with system rated Class A, or conform to BS 476 : Part 3 : 1987 with a minimum P60 rating for surface ignition and fire penetration time.
- F. Wind Uplift:
 1. The entire roof assembly and its roof ancillary components and flashing shall be installed to resist wind uplift of minimum 1.5KPa.
 2. Wind uplift pressure resistance at windward corners, roof edges, oversails, fascias, cappings, trims, drips, and around appurtenances shall be designed to minimum 1.5 times the design wind uplift values, unless higher design value is required in this specification.
- G. Asbestos, asbestos by-products, other known or banned hazardous materials, may not be used in any roof system material and ancillary products.
- H. The Manufacturer shall properly train, instruct and warn the Contractor in the proper use of roofing equipment.
- I. The roofing covering system shall be approved by BBA and shall not been penetrated within 1 hour and also achieve a minimum of designation EXT. F. AC when used on a non-combustible (concrete) substrate and tested in accordance with BS476 Part 3 (spread of flame).
- J. Thermal Performance Criteria

The thermal resistance of the complete roof system shall be minimum 0.027m²/k/W.

K. Solar and Weathering Performance Criteria

The Roof Covering System shall be approved by BBA and accelerated weathering tests confirm that satisfactory retention of physical properties is achieved and for an exposed system, with adequate repair and maintenance, indicate on expected life of at least 25 years.

Besides, when used in an inverted roof specification and subjected to normal service conditions, the system will provide an effective barrier to the transmission of liquid water for the design life of the roof in which it is incorporated.

L. Condensation

The Contractor shall design and construct the Roof Covering System in such a manner that no condensation takes place that has a deleterious effect on the Roof Covering System either on the internal surfaces or within the interstitial construction thickness. The Contractor shall ensure that the works conform to all aspects of the Specification taking into account all local environmental conditions prevailing at Site. The Contractor shall obtain any meteorological and climate data considered necessary in order to fulfil his obligations.

1. The Contractor shall provide condensation calculations demonstrating the suitability of his detailed design for a range of return periods, including but not limited to: 1 hour; 6 hours; 12 hours; 24 hours; 7 days; 30 days; and 60 days. Assumed internal temperature and humidity conditions at occupant level shall be:

Summer: Inside 24°C 65% relative humidity;

Winter: Inside 20°C 65% relative humidity.

Evaluation shall be on the basis of the relevant external surface being shaded and subject to an air velocity of 3.4 m/s during the summer and shaded and subject to an air velocity of 6.7 m/s in winter. Internal conditions shall be "still air" under both circumstances;

2. The Contractor shall design and construct the Roof System in such a manner that any surface condensation which may occur under conditions temporarily outside the limits stated above, subsequently evaporates without any deterioration of materials or fixings and provide calculations to demonstrate this.
3. The Contractor shall provide and install a non-combustible vapour barrier between the water proofing membrane and the roof insulation. The vapour barrier shall have taped joints and be sealed at the perimeter, and around upstands so as to prevent leakage. The final position of the vapour barrier(s) shall be determined by the Contractor in relation to the insulation layers and waterproof membrane;
4. The minimum tensile strength and puncture resistance of the vapor barrier are 156 lbs. force/ inch and 149 lbs. force, respectively according to ASTM E154; and
5. The vapor barrier shall conform to ASTM E1993-98 specification. Due to incorporate plasmatic core that helps maintain the maximum slab moisture transfer rate of 3 lbs. / 1,000 sq. ft / 24 hours.
6. The maximum water vapour permeance through the vapour barrier shall be 0.0011 Perms according to ASTM F1249, calibrated to ASTM E96.
7. Capillarity: The Contractor's detailed design shall eliminate water migration at joints and openings due to capillary action.
8. Impact and Abrasion Resistance: All surfaces shall resist heavy impacts from hand held objects and wind blown debris without any noticeable change to the

surface appearance, and the Roof Covering System shall resist abrasion from pedestrian traffic and cleaning operations without any noticeable change in surface appearance.

9. Movement Noise: The Contractor shall provide evidence to confirm that the intended form of assembly of the roof and the roof construction will not result in audible noise arising from relative movements of the construction. Internal ambient noise levels of NC55 in public areas and NC30 in IMAX cinema may be assumed for evaluation of this requirement. Audibility may be taken to occur at sound pressure levels 5dB below this rating curve.
10. Rigidity: The Contractor shall design and construct the Roof System so that it does not rattle, shake, creak or move in any way as to show signs of movement or produce audible noise.
11. Flashings: For flashings, 50mm lap joints secured with mechanical fasteners and the lapping shall be filled with silicone weather sealant. For gutters, sleeve of the same material and thickness of the gutters shall be mechanically fixed to the underside of the gutter joints. A minimum lapping of 150mm at each end of the sleeve shall be provided and shall be filled with silicone weather sealant.

1.05 ENVIRONMENTAL REQUIREMENTS

- A. The special precautions recommended by the Manufacturer shall be strictly followed where flammable and hazardous materials are involved or released. Hazardous waste materials shall be legally transported and disposed of in strict conformance to local regulations.
- B. The Contractor and Roof Manufacturer shall warrant that all products, materials and items are suitable, and wind uplift resistance and weatherproof seal can be maintained in the various climatic and wind conditions encountered without Roofing System failure and leaks.

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.07 SUBMITTALS

- A. Proposed sub-contractor specializing in installation of waterproofing and tank lining systems with ten (10) years minimum experience on projects of a similar size and complexity.
- B. Where reference is made to products from various manufacturers, submit a list of the proposed products and confirm compatibility between all elements/materials as appropriate.
- C. Alternatively a product from a single source supplier is permitted. Confirm compatibility between all elements/materials as appropriate.
- D. Shop Drawings :
 1. Indicate profiles, slopes, high and low points, all drain and penetration location, details, flashing, anchorage size and type of fasteners, and accessories.
 2. Include drawings, elevations and details where applicable. Plan Drawings shall be drawn at 1:20 scale. Detail drawings at 1:2 scale.
 3. Provide shop drawings by the Waterproofing and Tank Lining Manufacturer for the Consulting Engineers and the Architect's review and approval.

4. Show details at all penetrations, abutments, curb and wall transitions, drain and outlets, internal and exterior corners, and construction joint.
5. Details, locations and arrangement of tank lining, penetration flashing, construction joints and waterstops at all water retaining structures.

E. Samples :

1. Samples of 200 mm x 200 mm size for each tank liner, waterproof membrane, and coating specified herein. The cost of postage and transportation and all samples shall be the responsibility of the Contractor.
2. Sample of each lapped joint, pre-molded flashing piece, penetration boots, waterstops, all attached waterproofing and tank lining accessories.
3. The Contractor shall submit adequate samples for compatibility testing by the Testing Laboratory prior to any waterproofing and tank lining installation.
4. Samples of 600 mm x 600 mm (approx.) size for each roofing material, insulation, flashing, and joint material specified herein. The cost of postage and transportation and all samples shall be the responsibility of the Contractor.
5. Sample of each lapped joint, pre-moulded flashing piece, penetration boots, and all attached roof accessories.
6. The Contractor shall submit adequate samples for compatibility testing by Roof Manufacturer and Testing Laboratory prior to any Roof installation.

F. Testing and sampling reports as required in this section.

1.08 PRE-INSTALLATION CONFERENCE

- A. Record discussions of conference, decisions, and agreements reached, and furnish copy of recorded to each party attendant, and the Architect. Review foreseeable methods and procedures related to roofing, including the following:
1. Tour, inspect and discuss condition of substrate, preparation installation, and interface with other finish/material, penetrations and other preparatory Work performed by other building trades.
 2. Review waterproofing, penetration sealing, and tank lining requirements (Architectural and Consulting Engineers Drawing, Specifications and other Contract Documents).
 3. Review required submittals, both completed and yet to be completed.
 4. Review and finalize Construction Programme of Work related to this Work and verify availability of material, installer's personnel, equipment and facilities needed to make progress and avoid delays.
 5. Review required inspection, testing certifying and material usage accounting procedures.
 6. Review conditions, and procedures for coping with unfavourable conditions, including possibility of temporary protection.
 7. Testing, monitoring and sampling. Water quality analysis.

1.09 DELIVERY, STORAGE AND HANDLING

- A. All material shall be delivered in manufacturers' original containers, dry, undamaged, unopened packages, all clearly labelled with the Manufacturer's name, product identification, expiration date, and lot numbers intact.

- B. Materials shall be stored in strict accordance with the Manufacturer's printed instructions, copies of which will be furnished to the Architect.
- C. All lots and/or shipment of waterproofing material and accessories shall be tested within 30 days of receipt of materials according to the Manufacturer's procedure for determining field roofing material quality. Outdated materials may not be used. The Manufacturer's Technical Representative shall verify that all lots and/or shipments are from an authorized distributor and the roofing material and accessories are genuine products of the specified Waterproofing Material and Accessories Manufacturers.
- D. Store materials in weather protected environment, clear of ground and moisture. Protect insulation board from direct sunlight and flame/ignition sources. Care shall be taken to prevent contact of insulation board with solvent-based wood preservatives, coal tar and derivatives (e.g. creosote), paint thinner and other solvents (e.g. toluene) exposures. Store roll materials standing on end.
- E. Protect adjacent materials and surfaces against damage from all Roofing Work. Do not store materials on previously completed roof or deck. Do not stockpile materials on roof and decks so as to cause any overloading of any part of the Project Building Structures.
- F. Any special precautions recommended by the Manufacturer shall be strictly followed where flammable and hazardous materials are involved or released. Hazardous waste materials shall be legally transported and disposed of in strict conformance to local regulations and relevant BS standards.
- G. As recommended by manufacturer, and to comply with warranty requirements, store roofing materials at a temperature between 15.5oC and 16°C. If exposed to lower temperatures, restore materials to 15.5°C minimum temperature at least 24 hours before application by written procedures furnished by the Manufacturer.
- H. Storage and Handling of Roof Waterproof Membranes
 Deliver materials to Project Site in sealed, undamaged containers, or rolls on properly stored and wrapped pallets. Rolls must be stored on end. Do not double stack pallets of roofing membrane. Each container or pallet shall be identified with material name, date of manufacture and batch number.
 All material must be protected from prolonged exposure to sunlight by storing them under protected, weatherproof shelter. Shred material must be maintained in the original package at a minimum of 4°C and maximum of 60°C.
- I. Storage and Handling of Insulation Boards
 Deliver materials to the Project Site wrapped in polythene, undamaged, and on pallets. Each material pack shall bear the Manufacturer's label, date of manufacture, batch number, and BBA Agrément Certificate number.
 All material must be protected from prolonged exposure to sunlight by storing them under protected, weatherproof shelter.
 Insulation boards shall be prevented from contact with solvent-based wood preservatives, coal tar and derivatives (e.g. creosote), paint thinner and solvents (e.g. toluene, white spirit, etc.).
 Insulation boards shall not be exposed to naked flame or other ignition sources.

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

A. General Requirements:

1. Before commencing any Waterproofing and Tank Lining Works, the Contractor must demonstrate on Site that all adhesive and materials are fully compatible with each other and with the substances contained in the tanks and water retaining structures. No work shall be under taken when the surface moisture exceeds the permissible, as tested by the moisture testing equipment on Site. The spreading of adhesive over large areas before placing of membrane is forbidden. Keep record of the adhesive used and check against the agreed spreading rate of membrane. Bubbles formed in the membrane must be made good in accordance with the Manufacturer's technical installation literature.
2. Test Reports:
 - a. Submit a certified laboratory test report in accordance with Construction Product List A, Part 2, 1.10 by DIBt , include the following for the tests carried out in this Section : -
 - 1) To verify the compatibility between waterproofing and tile adhesive;
 - 2) Test results together with comparison with permitted parameters.
 - 3) Remarks and conclusions by the testing professionals.
 - b. Submit photographic records of the test arrangement, set-up and of the performance of the mock-ups, and constituent components of the same, during fabrication, assembly and under test. Submit dated photographic records one set each to the Architect and the Waterproofing and Tank Lining Manufacturers.
 - c. Provide copies of the agreed waterproofing and tank lining shop drawings for recording any modifications found necessary during and after the laboratory tests. The testing laboratory shall accurately and neatly record all changes, revisions, and modification etc., made to the shop drawings. At the completion of the testing, one copy each of the mark-up Drawings shall be given to the Contractor, who shall submit copies to the Architect, Consulting Engineers, the Waterproofing and Tank Lining Manufacturer.
 - d. Subsequently, and before any installation Work commence on the Project Site, the Contractor shall provide the Consulting Engineer and the Architect with one reproducible and two prints of the approved shop drawings incorporating the modifications made.
 - e. All retesting, and remedial or replacement work shall be paid by the Contractor and included in the Contract Sum.

- A. The Roof Manufacturer's Technical Representative shall conduct pre-installation and moisture test of all concrete roof slab substrate and cement/sand screeding to determine proper curing where required and specified herein.
- B. Flood testings of completed roof to ensure watertightness and proper drainage with no roof ponding.
- C. Conduct a pre-installation test of all roofing and ancillary materials for compatibility prior to actual roof work.
- D. Results of all tests shall be recorded in a Roof Field Test Log Book provided by the Roof Manufacturer. The Log Book shall be submitted as part of the Project Close-Out Submittal. The Log Book shall be protected by a waterproof covering and prevented from damage.
- E. Testing of Water Retaining Structures & Tanks
 - 1. Testing : Watertightness of structures:
 - a. Water retaining structures and tanks shall be tested for watertightness as stated herein. Each compartment of the tank or structure which incorporates division walls shall be tested separately with adjoining compartments empty; the complete structure shall also be tested.
 - b. The structure shall be filled with water at an approximately uniform rate not exceeding 2m depth in 24 hours to 150 mm below top of tank or structure. The water used for testing water retaining structures for potable or fresh water shall be fresh clean potable water. The Architect and Manufacturer's Technical Representative shall be notified before filling starts. The structure or each compartment of the structure being tested shall be kept full for 7 days before testing to allow for absorption.
 - c. After the period of absorption, the water shall be topped up to the specified level and the test shall begin. During testing, the oscillatory motion of the water surface shall be dampened. The test period shall be 7 days.
 - d. The equipment for recording water levels shall be installed in a temporary enclosure of minimum dimensions 2m x 2m x 2.5m high with a lockable door, the enclosure shall be located over stilling wells, manhole openings or other points of recording water levels. The temporary enclosure shall be removed on completion of the test. The equipment shall be calibrated before testing starts and at regular intervals reviewed without objection by the Architect and Manufacturers' Technical Representative, and shall be readable and accurate to 0.5mm.
 - e. The fall in water level in the tank or water retaining structure shall be measured at hourly intervals between 8 a.m. and 5 p.m. each day, the total fall shall be measured at the end of the test period.
 - f. Except as stated in (g) below, structures shall be emptied after completion of testing and maintained in a clean and dry condition. The Architect and Manufacturer's Technical Representative shall be notified before emptying starts.
 - g. The water used for the final tests on water retaining structures for potable or fresh water shall be retained in the structure and shall not be wasted or contaminated.

h. Compliance Criteria : Watertightness of structures

- 1) the total fall in water level at the end of the test period, after adjustment for evaporation and rainfall, shall not exceed 1/500 times the maximum specified depth of water in the test or 10 mm, whichever is less.
- 2) There shall be no leaks or damp patches visible on the surface of the structure, including any division walls, during or at the end of the test.

i. Non-Compliance : Watertightness of structures

- 1) If the result of any test for watertightness of a tank or water retaining structure does not comply with the specified requirements for the test, the Contractor shall investigate the reason. Remedial or replacement work reviewed without objection by the Architect, the Consulting Engineers, and the Waterproofing and Tank Lining Manufacturer shall be carried out and the structure shall be retested.
- 2) All retesting and remedial or replacement Work due to non-compliance shall be at the expense of the Contractor.

2. Testing : Water Sterility

a. Sample : Water sterilization

- 1) After the test for watertightness of a tank or water retaining structure for potable, or fresh water, has been completed, samples of the water in the structure shall be taken by the Testing Laboratory in the presence of the Architect, Contractor, and the Waterproofing and Tank Lining Manufacturer's Technical Representative. The number of samples and location of sampling shall be as directed by the Architect and the Water proofing and Tank Lining Manufacturer's Technical Representative.

b. Testing : Water sterilization

- 1) Each sample of water shall be tested to determine the bacteriological content and other harmful contaminants per BS 668, BS 6920 : Part 1 and Part 2 and all applicable local health and water quality regulations.

c. Compliance Criteria : Water sterilization

- 1) The results of tests for bacteriological content and other harmful contaminants of the water shall demonstrate that the structure has been adequately sterilised and exceed local health and water quality requirements.

F. Water test to roof and other wet and back of house areas; Infra-red scan test to roof:

1. After allowing waterproofing system installation to completely cure and set, flood test horizontal installations, in the presence of manufacturer's representative, with a minimum of 50 mm of water for at least 48 hours. Examine for, identify, and repair any leaks, then retest.
2. The test shall include temporary sealing of all outlets and if necessary construction of dams to compartmentalize large areas.
3. After flooding, all outlets blockages and dams shall be completely removed to drain the concerned areas. Do not permit any debris to enter into drainage pipework.
4. At a period between 24 and 48 hours from release of water provide an infra-red scan (roof areas only) carried out by an independent specialist contractor to establish if there has been any penetration through the membrane. Submit infra-red scanning report to the Architect.
5. Repair any defects in accordance with the manufacturer's instructions.

6. Compliance Criteria : There shall be no leaks or damp patches visible on the surface of the structure underneath, including any division walls, during or at the end of the test.
7. Non-Compliance :
 - d. If the result of any test for watertightness does not comply with the specified requirements for the test, the Contractor shall investigate the reason. Remedial or replacement work reviewed without objection by the Architect, the Consulting Engineers, and the Waterproofing and Tank Lining Manufacturer shall be carried out and the structure shall be retested.
 - e. All retesting and remedial or replacement Work due to non-compliance shall be at the expense of the Contractor.

G. Waterproofing Dry Film Thickness Test

1. The wet film thickness of each coat of waterproofing material shall be checked during application by measurements taken with a film gauge. Thickness shall be sufficient that when cured the dry film thickness shall be minimum recommended by the Manufacturer. The Manufacturer's Technical Representative shall verify dry film thickness (DFT) of all waterproofing materials at various locations on the Project Building.

1.12 WARRANTY

- A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.
- B. The Main Contractor and the Waterproofing, Roofing and Tank Lining Manufacturer / Supplier shall provide a (ten) 10 year warranty against material defects and workmanship for Work of this Section. The warranty shall commence at the date of Project Practical Completion.
- C. Nothing shall be done by the Contractor, which will void any Manufacturer's Warranty.

1.13 SPARE PARTS

- A. This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURERS/SUPPLIERS

- A. Subject to compliance with the Project Contract Document requirements, provide waterproofing, tanking lining, roofing system and required accessories from one or more of the following or equivalent approved by the Architect:
1. Tech Link Construction Engineering Limited
Unit 802, Laford Centre
838 Lai Chi Kok Road
Kowloon
Tel: (852) 3701 1817 / 9316 8906 Fax: (852) 3701 1818
Email: kenuskwong@techlinkcon.com.hk
 2. or approved equivalent

2.02 MATERIALS

- A. Provide and install all waterproof, tank lining and accessories listed in the below or equivalent approved by the Architect and as shown on the Contract Drawings, with ancillary material necessary for a proper, complete, weathertight and warrantable installation. (This indicates a partial list of principal items only).
- B. Joint sealants for the Work of this Section: Refer to Section 6.5.
- C. Refer to the Consulting Engineers' Contract Documents for site drainage, tanks, water-retaining structures, and service penetration requirements.
- D. Source Limitations for Waterproofing System: Obtain waterproofing materials from single source from single manufacturer.
- E. All waterproofing shall be proprietary product.
- F. "DP Drain Series – grade 200" high performance drainage composite panel (20mm thick) for Planter.
- G. "TREMCO TREMproof Amphibia" sheet-applied self-heals EPDM laminated waterproofing membrane comprised of 3 active layers hydro-reactive waterproofing membrane
- H. "BASF Conibase 2213" spray-applied fast cure Polyurethane (PU) waterproofing membrane system (min.2mm thick) for Internal Basement wall and floor as Basement internal tanking. The system components including min. 1mm thick "SCHOMBURG Asodur SG2" primer, min. 2mm thick "BASF Masterseal M800" spray-applied fast cure Polyurethane (PU) waterproofing membrane, and "BASF Mastertop P691" promoting primer sprinkled with silica sand to receive subsequently laid c/s render or screed.
- I. "EPRO e.stop HPL" (Formerly known as "EproSTOP-HPL") delayed hydration hydrophilic waterstop for construction joints.
- J. "DP 3000 Polyurea" (Formerly known as "DP Polyurea 3000") spray-applied fast cure Polyurea waterproofing membrane system (min. 1.5mm thick) for Podium Deck / Flat Roof. Used together with "DP adhesion promoting primer" sprinkled with silica sand to receive subsequently laid c/s render or screed if necessary.
- K. One layer of "DP 3000 SBS-SA" sheet-applied self-adhering waterproofing membrane (1.5mm thick), one layer of "DP 3000 SBS-4" sheet-applied SBS modified bitumen waterproofing membrane system (4mm thick) and one layer of "DP Geotextile" protection layer/separation layer (0.8mm thick) for Green Roof, Planter, Lawn.
- L. Two layers of "DP 3000 SBS-4" sheet-applied SBS modified bitumen waterproofing membrane system (4mm thick per layer) and one layer of "DP Geotextile" protection layer/separation layer (0.8mm thick) for flat roof and podium.

- M. "DP Brushcrete" polymer modified cementitious waterproofing coating in min. 2mm thick for water tank and in min. 1mm thick for all internal wet area including lavatory, plant room, pump room and refuse storage room and external wall with rendering finishes, etc.
- N. "BASF MasterProtect 300" (formerly known as "BASF Masterseal 300HS") cold liquid applied anti-carbonation and anti-chloride elastomeric waterproofing membrane without bitumen content in min. 0.5mm thick for external wall behind cladding.
- O. "SCHOMBURG Betocrete CL-170-P" (formerly known as "SCHOMBURG Betocrete C17(BV)") liquid form crystalline waterproof concrete plasticizer for lift pit and water tank, to provide waterproof concrete. Details should be referred to Particular Specification - Section 2.2 WATERPROOF CONCRETE.
- P. "PCI Seccoral 2K Rapid" trowel-liquid applied flexible and crack-bridging anti-efflorescence barrier with waterproofing function in min. 1.5mm thick for external wall with tiling finishes.
- Q. "BASF Masterseal 550 waterproofing membrane", min. 1 mm thick for the waterproofing system under raised floor
- R. "BASF Mastertop 1705 anti dust sealer", for the waterproofing system under raised floor. Details should be referred to Particular Specification - Section 8.9 SPECIAL COATING.
- S. "DP XPS X Series" polystyrene insulation board for in 50mm thick for flat roof and Podium Deck. Details should be referred to Particular Specification - Section 6.2 THERMAL PROTECTION INSULATION.
- T. Slip Sheets:
 - 1. Where there is a risk of plasticiser migration from the roof waterproofing membrane, a slip sheet of polypropylene or polyester felt, or similar material, must be interposed between the roof covering and the insulation board. The slip sheet shall be of a type compatible with both the insulation board and the roof waterproofing membrane. The Roof Manufacturer shall confirm by test whether slip sheets are required.
 - 2. Test report shall be submitted to the Architect.
- U. Vapour Barriers: Vapor seal with plasmatic core. Comply with ASTM E1993-98, Water Vapor Permeance Rating as low as 0.0011 perms according to ASTM F1249 calibrated to ASTM E96, and compatible to all roofing system materials.

2.03 ANCILLARY MATERIALS

- A. All ancillary material shall be from the same waterproofing or tank lining manufacturers and they shall be compatible with each other.
- B. Other required ancillary materials and product as indicated in the approved shop drawings, waterproofing and tank lining manufacturer's written installation instructions.

2.04 PERFORMANCE REQUIREMENT

- A. "DP Drain Series – grade 200" high performance drainage composite panel (20mm thick) for Planter.

Description:

3D Drainage sheet is made of high molecular, produced by extrusion and adsorption. The tri-dimensional space can drain gas and fluid. It is a multi-functional material, can

form channels for water and gas easily, which is sound insulation and vibration isolation.

Sheet thickness 0.8 - 2.0 mm, dimple height is 20 mm thick, width is 2~2.5m, sheet materials are HDPE, EVA, etc., length according to user's requirements.

Other products submitted by the contractor may be considered for incorporation subject to compliance with the performance and properties, and approval by the Architect. Minimum Performance of drainage panel:

- B. "BASF Conibase 2213" spray-applied fast cure Polyurethane (PU) waterproofing membrane system (min.2mm thick) for Internal Basement wall and floor as Basement internal tanking.

As Basement internal tanking, the system components including min. 1mm thick "SCHOMBURG Asodur SG2" primer, min. 2mm thick "BASF Masterseal M800" spray-applied fast cure Polyurethane (PU) waterproofing membrane, and "BASF Mastertop P691" promoting primer sprinkled with silica sand to receive subsequently laid c/s render or screed.

Other products submitted by the contractor may be considered for incorporation subject to compliance with the performance and properties, and approval by the Architect. The proprietary Basement external tanking and Basement internal tanking shall meet the following minimum performance accordingly:

- i. Minimum performance of "SCHOMBURG Asodur SG2" (formerly known as "SCHOMBURG Indufloor IB1240") (min. 1mm thick) sprinkled w/ silica sand:
- The manufacturer must have the quality standards of EN ISO 9001;

Item		Index		
		Sheet thickness < 0.8mm	Sheet thickness 0.8mm-1.0mm	Sheet thickness ≥ 1.0mm
Tensile strength ≥		40 N/mm	56 N/mm	72 N/mm
Tensile elongation ≥		25 %	35 %	50 %
Compressive property	Compressive strength ≥	100 KPA	150 KPA	400 KPA
	Appearance after 50% compressure	No broken		
Drainage capacity/cm2 ≥		30		
Air aging (80°C X168h)	Tensile strength retain ≥	80		
	Tensile elongation retain ≥	70		
Anti-acid and alkaline (saturated Ca(OH) ₂ solution 23°C X168h)	Tensile strength retain ≥	80		
	Tensile elongation retain ≥	80		

- A unique 2-component, moisture tolerant, extremely high-density epoxy based product, which prevents the passage of water vapour and moisture through slabs on or below grade, thus eliminating delamination of adhesives, floor coverings and coatings. It also prevents capillary infiltration of oil or other chemicals from the ground and can be used to treat oil-contaminated slabs;
- Reduce water vapour emission levels of up to 25 lbs/1000 ft²·24 hrs (5 g/m²·hrs) to well below accepted levels for subsequent application of flooring systems such as epoxies, vinyl tiles, sheet vinyl, carpets, wood flooring, etc;
- Resistant to chemicals rising from the soil with an excellent bonding on wet concrete surfaces. It is used as a primer for oil contaminated, but cleaned concrete surfaces and for the priming of still damp concrete or screed surfaces which will be coated with epoxy flooring systems;
- Other properties are shown as below:

Typical properties	Result
Basis	two component epoxy resin
Colour	light grey
Viscosity	70 s in a 4 mm DIN standard cup
Density	1.86 g/cm ³
Mixture ratio	100:12 parts by weight
Pot life	60 minutes at +23°C 30 minutes at +30°C
Minimum cure temperature	+10°C (slow cure)
Cured	after 7 days at +23°C
Foot traffic	after 12 hrs. at +23°C
Consumption	min. 600 g/m ²
Compressive strength	80 N/mm ²
Flexural strength	30 N/mm ²
Tensile adhesive strength	Dry surface: > 2.0 N/mm ² (EN 246324) Wet surface : > 1.5 N/mm ² (EN 246324) ≥ 2.5 N / mm ²
Adhesion to	New concrete 110 psi 5 days old (0.8 Mpa) Concrete, moist 550 psi 28 days old (3.8 Mpa) Concrete, dry 580 psi 28 days old (4.0 Mpa)
Working time	1 hrs*)
Open to foot traffic	12 hrs*)
Recoat time	12 hrs*)
Full strength	7 days*)
Radon Barrier	< NWG m ² / s (0.6mm thick)
Moisture Vapor	0.09 g/m ² / 24hrs average or less, under

Typical properties	Result
Transmission	901µm thickness
Sd [H2O]	230m or more, under 901µm thickness
*) at an ambient temperature of 73°F (23°C) at 60% relative air humidity	
Abrasion resistance	≤209mg (ASTM D 4060:01)
Capillary Water Absorption and Water Permeability	W <0.1 kg/m ² x h0.5 (DIN EN1504-2:2004)
Reaction to Fire	Class E (DIN EN 1504-2:2004)
Water Vapor Permeability	Class III (DIN EN 1504-2:2004)
Moisture Vapor Transmission	≤0.08g/m ² /24hours (DIN 53122 T1 free film), under 677 µm thickness
Water vapor diffusion resistance	µ≥738,000 (DIN 53122 T1 free film), under 677µm thickness
Sd [H2O]	≥500m (DIN 53122 T1 free film), under 677µm thickness
Compressive strength	>87 N/mm ²
Flexural strength	>40N/mm ²
Adhesive strength	≥2.5N/mm ² (ASTM D4541:02)

- C. "TREMCO TREMproof Amphibia" sheet-applied self-heals EPDM laminated waterproofing membrane comprised of 3 active layers hydro-reactive waterproofing membrane.

External Tanking System shall be composite membrane of min.1.3mm thickness consisting of an EPDM watertight barrier layer, a hydro-reactive polymer core layer, an EPDM derivative/polymer hydro-reactive layer, and a non-woven polyethylene geotextile facing; or approved equivalent together with accessory materials including but not limited to Termination Bar, Tapes, Penetration Waterstops, Termination Seals, and the like recommended by waterproofing manufacturer for intended use and compatible with sheet waterproofing in accordance and in compliance with the Manufacturer's printed data, drawings and instructions.

1. Physical Properties

- a. Composite membrane
- b. General: Waterproofing system shall be capable of performing as a continuous watertight installation. Waterproofing shall accommodate normal substrate movement, construction material transitions, opening transitions, penetrations, and perimeter conditions without resultant moisture deterioration.
- c. Compatibility: Provide waterproofing system materials that are compatible with adjacent materials under conditions of service and substrates on which product is applied, as recommended by waterproofing manufacturer based on testing and field experience.
- d. Multi-Layer Basement Sheet Waterproofing: Water-reactive pre-applied membrane, self-fastening to the concrete, suitable to waterproof reinforcing concrete structures (raft foundations) as similar with equal or superior performances. The membrane shall be composed of a coextruded

polymeric multilayer sheet, with the following four layers with unique features and functionality:

- 1) The external waterproofing EPDM based water-tight barrier.
 - 2) The central self-sealing core layer (a blend of synthetic rubber – ethylene propylene diene monomer EPDM, mineral & reinforcing fillers, a proprietary water-swellable polymer and processing aids) which seals autonomously any punctures such as nails & staples and helps seal penetrations.
 - 3) The third active layer (a different mix of synthetic rubber – ethylene propylene diene monomer EPDM, mineral & reinforcing fillers, a proprietary water-swellable polymer and processing aids) which self-seals the overlaps and helps avoid any chance of lateral migration of water between the interface of concrete and the membrane.
 - 4) The finish of the internal layer (Active layer) shall be a proper non-woven textile suitable to allow the self-fastening to concrete.
- e. The waterproof membrane shall be made of EPDM water-tight barrier with a combination of flexible and active materials (as above mentioned) such that all critical areas such as bends, corners, up/down turns and tightly accessible areas can be easily applied to ensure a continuous waterproof envelop.
 - f. Any materials which is not as flexible as EPDM such as HDPE, PVC or similar should not be used. Besides, the membrane should be fully bonded with the concrete surface by means of a non-woven textile or similar mechanical bonds. Any chemical bonding mechanism, such as pressure sensitive adhesive, whose effectiveness could be compromised due to weather or dust should be avoided.
 - g. The substructure and superstructure waterproofing works for basement floor and wall shall be done by the same authorized waterproofing applicator and by the waterproofing system supplier to ensure the continuity and quality of the waterproofing installation work for the water tightness of the basement structure.
 - h. The waterproofing system shall allow easy insertion of dowel bars through the self-sealing membrane, and be easy to staple over vertical formwork or fix mechanically as any insertion through the membrane will self-repair.
 - i. The waterproofing system shall be resistant to aggressive natural chemicals contaminated ground water or soil, provide protection against roots penetration, less affected by natural elements like rain, dust and wind
 - j. The waterproofing system shall provide protection against radon and methane gas.
 - k. The waterproofing system shall be work-friendly. It shall allow repairing in patches, and be easy to bend and apply over turns and difficult access area.
2. Additional Performance Requirements:

Property	Test Method	Result
Thickness	Not including non-woven fabric layer	1.3mm
Mass per unit area	EN 1849-2	max. 1.43 kg/m ²
Water Tightness	EN 1928 B (700 KPa/24 hours)	No passage
Resistance to chemical agents and water tightness	EN 1847 (CaOH ₂ -28 days) EN 1928 B (700 KPa/24 hours)	Test Passed
Resistance to accelerated	EN 1296 (12 weeks 70°C)	Test Passed

weathering and water tightness		EN 1928 B (700 KPa/24 hours)	
Tear Strength-Longitudinal		ASTM D624-00(2012)	>52 N/mm (average)
Tear Strength-Transverse		ASTM D624-00(2012)	>47 N/mm
Tensile strength-Longitudinal		ASTM D412:2016 (Method A)	>4.6 MPa
Tensile strength-Transverse		ASTM D412:2016 (Method A)	>3.6 MPa
Elongation at Break-Longitudinal		ASTM D412:2016 (Method A)	>850% (average)
		EN12311-2	>250%
Elongation at Break-Transverse		ASTM D412:2016 (Method A)	>1000% (average)
		EN12311-2	>250%
Fire Resistance		EN 13501-2	Class E
Puncture Resistance		ASTM E154/E154M-08a (2013) e1	>750 N (average)
Low Temperature Flexibility : Longitudinal		ASTM D 1970/D1970M-17a	Pass
Low Temperature Flexibility: Transverse		ASTM D 1970/D1970M-17a	Pass
Hydrostatic Pressure Resistance		ASTM D5385 / D5385M-93 (2014) e1	Pass at 7 bars
Lateral Water Migration		ASTM D 5385 Mod.	Pass at 7 bars
Watertightness at Seams		EN 1928-B Mod	Pass at 7 bars
Water Vapour Permeance		ASTM E96-12 (Desiccant Method)	1. <0.6 ng/Pa*s*m2 (average) 2. <0.003x10-2 g / hour*m2 (average)
Peel Adhesion to Concrete		ASTM D 903:1998 (2017)	>1.2 N/mm (average)
Low Temperature Crack Bridging		ASTM C836, -23°C	Pass
Cutting resistant		Internal test by using cutter under 1.5 bars water pressure	No leakage
Crack bridging and water-tightness after the test	a. No. of cycles: 10,000 times b. Dynamic crack movement: 1.75 to 3.75mm c. Water-tightness test after the 10,000 times crack cycles	EN14224:2010 & 1.5 bars water pressure for 7 days	No leakage
Lap Peel Adhesion		ASTM D1876	>4 N/mm (average)
Lap Shear Strength		ASTM C961	>125 KN/m2 (average)
Resistance to root penetration		EN14416	Passed
BBA approval		BBA	Yes
CE Certification		EN13967:2012	Yes

3. Accessory Materials

- a. General: Furnish accessory materials recommended by waterproofing manufacturer for intended use and compatible with sheet waterproofing.
- b. The waterproofing in all vertical applications should be mechanically fastened with the substrate to ensure a vertical levelled surface (without sagging / folding) is achieved to provide an optimum confinement for the membrane between the newly placed concrete and the substrate.
- c. Tapes for Floor Overlap: Waterproofing manufacturer's recommended tape for joints between membranes.
 - 1) Safety Overlap Tape: double-sided, synthetic rubber composite membrane tape, with a butyl adhesive on one side and a fleece material facing.
- d. Wall Overlap: Waterproofing manufacturer's recommended sealant for joints between membranes.
- e. Penetration and External/Internal Corners Waterstops: Hydro-swelling Mastic or similar
- f. Basis of Design Product – AKTI-VO201

ii. Minimum performance of "BASF Masterseal M800" spray-applied fast cure Polyurethane (PU) waterproofing membrane:

The waterproof membrane shall be a fast cure spray applied two-component solvent free elastic waterproof coating and contains no fillers in any part of the materials. The cured liquid polyurethane membrane shall not be less than 2mm thick, applied in a sprayed single coat and capable of accepting light foot traffic after three minutes. If small application areas where spray application is not feasible, its hand applied version "BASF Masterseal M860" is acceptable.

The cured, waterproof membrane shall be root and chemically resistant, particularly to the effects of water, sea and waste water, as well as a variety of alkalis, diluted acids, mineral oils, lubricants and fuels. Physical Properties are:

- The cured membrane shall have root penetration resistance and be proved by DIN 4062.
- The waterproofing membrane shall carry the green label certification for Singapore.
- The waterproofing membrane shall be fast setting (tack free time approx. 30 seconds) and tough enough to receive foot traffic in within 180 seconds after spraying and should have no defects.
- It shall be proved by resisting to negative water pressure of minimum 10 bar (100m water head) according to DIN 1048.
- The tensile strength of waterproofing membrane shall be not less than 10MPa when tested in accordance with a recognized international standard having been reviewed without objection by the engineer.
- The waterproofing membrane shall carry the potable water certification from WRC-NSF and have no negative effects to the water. It shall also comply with the requirements of BS 6920, Part 1, clause 8.
- It shall be capable of bridging cracks up to 6 mm wide which develop after the installation of the membrane and shall have an elongation of break of not less than

400% when tested in accordance with a recognized international standard having been reviewed without objection by the engineer.

- The Coniroof system shall be one of the ASD approved proprietary waterproofing membrane roofing materials of 'Type BII Liquid Applied Roofing Materials with Self Finishes' under the List of Accepted Materials for Roofing Systems stated in Annex 1, Clause 5 of ASD Operational Instruction No. 11/2006 as attached in Appendix 1 or any latest version issued by ASD.
- The waterproofing membrane shall meet the fire exposure test according to BS 476 part 3 and is classified as "Ext.F.AA" on cement board and "Ext.F.AC" on concrete.
- Subject to adequate repair and maintenance provisions satisfactory retention of physical properties of the system shall be achieved for 10 years.
- The installation of the waterproofing shall be carried out by an approved applicator approved by the manufacturer, trained to use the specialist equipment and in full accordance with the technical data sheet and application instructions according to Swiss Standard ZTV-BEL B3/87.
- Withstand the high temperature during laying of hot poured asphalt (approx. 240°C)
- It shall allow the diffusion of water vapour to prevent build up of pressure between the membrane and the substrate.

Property	Characteristic Value	Testing Method
Description	Solvent free two component spray applied waterproofing membrane	--
CE-Marking	EN 1504-2: ZA.1d, ZA.1e, ZA.1f and ZA.1g	EN1504-2
Roofing Materials	Type BII Liquid Applied Roofing Materials with Self Finishes	ASD approved
Withstand temperature during laying of hot poured asphalt (approx. 240°C)	YES	---
Cadmium (Cd), Lead (Pb), Chromium (Cr) & Mercury (Hg)	EPA 3052 & EPA 6010B	Not Detected
Tensile Adhesion Bond Test	>4MPa	ASTM D4541
Adhesion strength after 1000 hrs (41.7 days) QUV exposure	>2.60 N / mm ²	ASTM D4541 & G154
Adhesion to steel	>10 MPa	Conica
Resistance to aggregate Chisel impact test	no charge no charge	Conica
Ready for foot traffic	Max. 60 seconds	---

Solid Content	100%	---
Hardness	73 average	ASTM D2240
	78	ASTM D2240
Shore A hardness	80	---
Median Tensile Strength	9.4 – 11.9 N/mm ² average (1.5 – 2mm thick, 1.3 & 7 days); shall not be over 16 N / mm ²	ASTM D415
Tensile strength (sprayed)	10.0 N/mm ²	DIN 53504
	≥11N / mm ² average (2 mm thk, 3 days)	ASTM D412
Tensile strength Heat treatment Exposure treatment Alkali treatment Acid treatment	102 kg f / cm ² (Passed) 92 kg f / cm ² (Passed) 93 kg f / cm ² (Passed) 97 kg f / cm ² (Passed)	JIS A 6021 JIS A 6021 JIS A 6021 JIS A 6021
Elongation (sprayed)	400%	DIN 53504
	≥550% average (2 mm thk, 3 days)	ASTM D412
Median Elongation	530 – 580% average (1.5 – 2mm thick, 1.3 & 7 days)	ASTM D412
Elongation at break Heat treatment Exposure treatment Alkali treatment Acid treatment	700% (Passed) 498% (Passed) 677% (Passed) 625% (Passed)	JIS A 6021 JIS A 6021 JIS A 6021 JIS A 6021
Tear strength (sprayed)	18 N/mm ²	DIN 53515
	≥44.65N/mm average	DIN 53363
Tear Strength	40 – 47 N/mm	DIN 53363
Water vapour transmission rate	1. 19 g/m ² • 24 hrs (1.5mm thk.)	1. BS3177 (25°C / 75% RH)
	2. 0.11 g/m ² • 24 hrs (1.61 - 1.76mm thk.) 3. >5.6 g/m ² • 24 hrs (2.1mm thk.)	2. ASTM E96 method B 3. ASTM E96 method B
Water vapour permeability	800 ym	DIN 52615
	800 average	DIN EN7783-1

Static crack bridging at -20°C	4.5mm (-20°C)	Munchen Technical University
Water absorption	7.6 x 10 ⁻³ kg / m ²	DIN EN 14224
	3.99%	EN ISO 62 : 08
Dynamic crack bridging under exposure / load of melted asphalt	Passed 1000 cycles 0.2mm crack at -20°C	ZTV – SIB90
	Passed 1000 cycles of 0.25 – 0.75mm (-20°C)	DIN EN14224
Heat aging at 70°C	28 days, no change	ZTV-SIB 90
Crack Bridging ability	> 6 mm (22°C)	Munchen Technical University
	≥ 5 mm	ASTM C836
Static crack bridging at -20°C	4.5mm (-20°C)	Munchen Technical University
Taber Abrasion Resistance	< 40 mg	ASTM D4060
	22.5mg (CS-17, 500cycles)	ASTM D4060
Watertightness to negative water pressure	Resistant to 10 bar (100m water head) back-pressure	DIN 1048
Root penetration resistance	Pass	DIN 4062
Potable Water Safety	Pass	BS 6920
Reaction to Fire	Ext 'F.AA' on cement board Ext 'F.AC' on concrete	BS 476: Part 3
	1. None of the samples the flame tip reaches the mark of measurement. 2. None of the tests shows burning droplets	EN ISO 11925-2
Reaction to fire classification	E - freely suspended	EN 13501-1
Electric resistance a. Surface resistance (MΩ) b. Volume resistance (MΩ • cm)	4.2 x 10 ⁶ 2.1 x 10 ⁵	ASTM D257
Chemical resistance to Engine oil (23°C, 7 days) Gasoline (23°C, 7 days) Diesel (23°C, 7 days) 10% CaCl ₂ (23°C, 7 days) 10% NaCl (23°C, 7 days)	No change on surface No change on surface No change on surface No change on surface No change on surface	ASTM D543
Chemical resistance to Acetic acid 10%	Resistant	---

5% 2% Ammonia 5% Boric acid 4% Caustic potash solution 10% Citric acid 10% Disinfectants Distilled water Ethanol 10% Formaldehyde Formic acid 10% 5% 2% Glycerine Hydrochloric acid 10% Hydrogen peroxide 10% Kerosene Lactic acid 25% Phosphoric acid 25% Potassium hydroxide 10% Salt solution 10% Soda lye 50% 40% 5%		
Soda solution 20% Sodium carbonate 20% Sodium chloride 10% Sodium hydroxide 50% 40% 25% Sugar solution 30% Sulphuric acid 25% 10% Tannic and solution Tap water		

Chemical Resistance, after 7-day immersion 1. 10% HCl (hydrochloric acid) 2. 10% CH ₃ COOH (acetic acid) 3. 20% Na ₂ CO ₃ (sodium carbonate) 4. 10% NaCl (sodium chloride) 5. 10% NaOH (sodium hydroxide) 6. Sea water 7. Diesel 8. 10% H ₂ SO ₄ (sulphuric acid) 9. 10% HNO ₃ (nitric acid) 10. 10% H ₃ PO ₄ (phosphoric acid) 11. Hydraulic oil 12. Brake fluid	No disintegration, blistering, swelling, peeling or cracking	ASTM C868 : 2008 (7 days full immersion)
Expected service life after accelerated weathering tests for an exposed system	At least 25 years	BBA
Volatile Organic Compound (VOC) Total Halogenated Organic Solvet Total Aromatic Organic Solvent Epichlorohydrin	0.01 g/ltr Not Detected 0.0003 Not Detected	Headspace GC-MSD Headspace GC-MSD Headspace GC-MSD Headspace GC-MSD
Green Label from Singapore	Singapore Green Labelling Scheme (SGLS)	Yes
Formaldehyde	Not Detected	High Performance Liquid Chromatography

Resistant to Ballast Penetration	Can withstand the stresses imposed during the pulsating load test with test loads equivalent to an axle load of 280 kN, over 2.7×10^6 cycles and without perforation	Munchen Technical University
Ballast Penetration	Pass	German Federal Railways test
Ozone resistance	Passed	Conica
Tack free time Ready for foot traffic	---	30 sec. 180 sec.
Impact Resistance	203 N/mm	ASTM G14:04
Resistance to Puncture	340.50 N	ASTM E154:1999(05)
P621 Resistant to Moisture Testing including bond strength	---	DS/EN 13892-2
Hydrostatic Pressure Test	5 bars	DIN 16726

- iii. Minimum performance of “BASF Mastertop P691” promoting primer sprinkled with silica sand to receive subsequently laid c/s render or screed:

It is a single component, moisture curing polyurethane (no epoxy base shall be used) adhesion promoting primer fully sprinkled with silica sand. It shall be used if cementitious render/screed to be directly applied on top of “BASF Masterseal M800” (or hand applied version “BASF Masterseal M860”) waterproofing membrane, to increase the adhesion strength with the subsequently applied cementitious layer.

Physical properties:

Property		Characteristic Value
Chemical Base		PU
Density		1.03 g/cm³
Viscosity		110 mPas
Re-coating interval	at 23 °C / 50 % r.h.	min. 1 h.
		max. 24 h.
	at 23 °C / 60 % r.h.	min. 2 h.
		max. 36 h.
Substrate and ambient temperatures		min. 5 °C
		max. 30 °C
Permissible relative humidity		min. 40%
		max. 90%

D. “EPRO EproSTOP-HPL” delayed hydration hydrophilic waterstop for construction joints

Other products submitted by the contractor may be considered for incorporation subject to compliance with the performance and properties, and approval by the Architect. The proprietary delayed hydration hydrophilic waterstop for construction joints shall meet the following minimum performance:

- Basic Use: ‘Eprostop-HPL’ is designed to self joints in many forms of concrete and concrete construction such as cold joints, penetrations, etc. when exposed to moisture thus, maintaining water tight integrity.
- Composition: ‘Eprostop-HPL’ is a rubber based product that has been formulated with special hydrophilic compounds that are intended to expand in a controlled when exposed to moisture.
- Benefits :
 - i. Does not over expand which can cause self-deterioration.
 - ii. Does not over stress adjoining substrate material.
 - iii. Excellent resistance to hydration / dehydration cycle damage.
- Other Physical Properties :

Property	Testing Standards	Result
Base	-	Rubber (no bentonite inside)
Specific Gravity	ASTM D71	1.35 + 5
Hydrocarbon Content	ASTM D4	47% min.
Volatile Matter	ASTM D6	1% max.
Eprostop Penetration, cone 77°F, 150gm 5 sec.	ASTM D217	40 + 5
Delayed Hydration Type	-	Yes (Need to fully immerse into water for at least 3 to 4 days before swelling starts)

E. “DP 3000 Polyurea” (Formerly known as “DP Polyurea 3000”) spray-applied fast cure Polyurea waterproofing membrane system (min. 1.5mm thick) for area as specified in waterproofing schedule and demarcation drawings including but not limited to Podium Deck / Flat Roof. Used together with “DP adhesion promoting primer” sprinkled with silica sand to receive subsequently laid c/s render or screed if necessary

Other products submitted by the contractor may be considered for incorporation subject to compliance with the performance and properties, and approval by the Architect. The proprietary spray-applied fast cure Polyurea waterproofing membrane system shall meet the following minimum performance:

- i. The general purpose primer of Polyurea waterproofing membrane for dry concrete substrate:

Test Items		Minimum Performance	Test Method
Tensile Strength (MPa)		34.0	ASTM C307-18
Absorption (%)		0.4	ASTM C413-18
Compressive Strength (MPa)		90.0	ASTM C579-01
Modulus of Elasticity (MPa) (Flexural Strength)		79.0	ASTM C580-18
Tensile Properties	Tensile Strength (MPa)	65.0	ASTM D638-14 (V=5mm/min)
Flexural Properties	Flexural S(MPa)	69.0	ASTM D790-17 Procedure AI
	Flexural Modulus (MPa)	2719.6	ASTM D790-17 Procedure AI
Durometer Hardness (Type D 1 Sec)		83	ASTM D2240-15 (2021)
Abrasion Resistance (g) (CS-17,500g, 50 cycle)		0.0024	ASTM D4060-19

- ii. Polyurea waterproofing membrane:

The waterproof membrane shall be a fast cure spray applied two-component polyurea membrane shall not be less than 1.5mm thick, applied in a sprayed single coat and capable of accepting light foot traffic after three minutes.

The waterproofing system shall be a multi-layered system comprising a sealer/primer applied on substrates and waterproofing membrane, that prevents the passage of water. The sealer / primer shall be supplied by the same waterproofing manufacturer.

The cured, waterproof membrane shall be root and chemically resistant, particularly to the effects of water, sea and waste water, as well as a variety of alkalis, diluted acids, mineral oils, lubricants and fuels. Physical Properties are:

Properties	Minimum Performance	Testing Methods
EOTA certified material for liquid applied roof waterproofing based on pure polyurea	Yes. EOTA certified material for liquid applied roof waterproofing based on pure polyurea	Guideline for European Technical Approval (ETAG)
Tensile Strength -Test Temperature 23°C	≥ 22.5 N/mm ²	CNS 8644 (2014)

-Test Temperature - 20°C	≥ 29.3 N/mm ²	
-Test Temperature 60°C	≥ 19.4 N/mm ²	
Tensile Stress at Elongation	a. 100%: ≥10.3 N/mm ² b. 200%: ≥14.0 N/mm ² c. 300%: ≥19.0 N/mm ²	ASTM D412-16a
Elongation at 23°C	≥ 435%	CNS 8644 (2014)
Tear Strength	≥ 110 N/mm	CNS 13333 (1993)
Hardness	a. Shore D ≥ 51 b. Shore A ≥ 95	ASTM D2240-05 (2010)
Abrasion Resistance	≤ 0.01g (CS-17, 1000g, 1000 r.p.m.)	CNS 10757 (1995)
Dynamic Crack Bridging	13.5 mm no cracking after 100 cycles	BS EN1062-7:2004 Method B
Crack Bridging	No cracking at minimum 8mm stretching stroke	ASTM C1305-95 MOD
Efflorescence Test	No efflorescence	JC/T 1024-2007 after 21 cycles
Puncture Resistance	a. Maximum Force: ≥ 2090N average b. Deflection ≥ 78 mm average	ASTM E154-08a (2013)
Adhesion to concrete	≥ 4.1 MPa	ASTM D4541-09
Adhesion to overlaid c/s screed (with adhesion promoting primer on top of waterproofing membrane)	≥ 1.3 MPa	BS EN 1348
Overlap adhesion between 2 coats of waterproofing membrane	≥ 6 MPa adhesion between 2 coats of waterproofing membrane	BS EN ISO 4624-16 Clause 8.4.2
Mandrel Bend Test	No cracking at minimum 3mm Mandrel Diameter	ASTM D522-93a
Suitable for use in contact with water for human consumption	Pass	BS6920 Part 1:2000
WRAS Certificate	Approved Material	WRAS
Green Product Approval	“Platinum” rating	Hong Kong CIC Green Product Certification
Green Product	Certificated Green Product	Singapore Green

Approval		Labelling Scheme
Ozone Depleting Substance	Not detected	ISO 11890-2
Plasticizers 1. BBP 2. DBP 3. DEHP 4. DINF 5. DIDP 6. DNOP	Not detected	CPC-CH-C1001-09.4
Formaldehyde content	Not detected	1. High Performance Liquid chromatography 2. UV-Vis
Mercury, Lead, Cadmium and Chromium	Mercury (hg): not detected Lead (Pb): not detected Cadmium (Cr): not detected Chromium (m): not detected	EPA 3052 Microwave acid digestion, EPA 6010B: ICP (Inductively coupled Plasma)
Flash point	Meet SGLS 032 Criteria >61°C	ASTM D3828-16a (Procedure B)
Total Halogenated solvent, Total Aromatic Organic solvent, Epichlorohydrin and N-Methyl Pyrrolidinone	Halogenated solvent: not detected Aromatic solvent: not detected Epichlorohydrin: not detected Epichlorohydrin and N-Methyl Pyrrolidinone: not detected	ISO 11890-2
Pb, Sb, As, Ba, Cd, Cr, Hg, Se etc. heavy metal	Not detect	CNS 4797-2
VOC test	Not detect	1. GC/MC testing 2. ISO 11890-2
Solids by Volume	100%	-
Cure time before foot traffic	≤60 sec. (25°C)	-
Water Vapor Permeability	a. Permeance: ≤0.99 ng / Pa·s·m ² b. Vapor Transmission: ≤0.005 g / hr·m ²	ASTM E96-12 (Procedure B-inverted water method)
Resistance to Water Vapor	μ = 2279	EN 1931
Water Absorption	≤ 0.50%	ASTM D570-98

		(2010)									
Chemical Resistance 20% HCl 50% NaOH 100% H ₂ SO ₄	No visible change on the appearance	ASTM D543-06									
Root Resistance	No root penetration	DD CEN / TS 14416:2014									
Positive Water Pressure Resistance	16 bars	DIN 1048 Clause 7.6									
Negative Water Pressure Resistance	12 bars	DIN 1048 Clause 7.6									
Watertightness	Watertight	EOTA TR-003									
External fire performance	Classification B _{roof} (t1)	EN 13501-5									
Statement of dangerous substances	Does not contain and release any dangerous substance	EOTA TR 034									
Resistance to wind loads	Pass (> 50 kPa)	EOTA TR-4									
Resistance to dynamic indentation	Resistance Level: I ₄	EOTA TR-6									
Resistance to static indentation	<table border="1"> <thead> <tr> <th>Support</th><th>Load (N)</th><th>Resistance level</th></tr> </thead> <tbody> <tr> <td>Steel</td><td>250</td><td>L4</td></tr> <tr> <td>Extruded polystyrene</td><td>250</td><td>L4</td></tr> </tbody> </table>	Support	Load (N)	Resistance level	Steel	250	L4	Extruded polystyrene	250	L4	EOTA TR-7
Support	Load (N)	Resistance level									
Steel	250	L4									
Extruded polystyrene	250	L4									
Resistance to fatigue movement (1000 cycles)	Pass	EOTA TR-8									

Resistance to high temperatures effects	Temp °C	Support	Load (N)	Static Indentation Resistance Level	EOTA
	60°	Steel	250	L4	
		Extruded polystyrene	250	L4	
	90°	Steel	250	L4	
		Extruded polystyrene	250	L4	
Resistance to heat ageing	The samples are exposed to 80°C during 200 days				EOTA TR-11
	Properties		Values		
	Fatigue movement		Pass		
	Dynamic Indentation (-20°C)		14		
	Tensile strength (MPa) (EN ISO 527-3) (initial / ageing)		23 / 17		
	Tensile elongation (%) (EN ISO 527-3) (initial / ageing)		315 / 372		
Resistance to UV-radiation in the presence of moisture	The sample are exposed 5,000 hours to UV-radiation				EOTA TR-10
	Properties		Values		
	Dynamic Indentation (-10°C)		I4		
	Tensile strength (MPa) (Type 2) (EN ISO 527-3) (initial / ageing0		23 / 17		
	Tensile elongation (%) (EN ISO 527-3) (initial / ageing)		315 / 372		
Resistance to wind loads	Pass (>50 kPa)				EOTA

Resistance to plant roots	Resistance to plant roofs	EN 13948
Safety in use (BWR 4) Slipperiness	Rd = 50	ENV 12633:2003 Annex A
Related aspects of serviceability: Effect of weather conditions	Pass	EOTA
Related aspects of serviceability: Effect of day joints	Pass	EOTA

iii. Adhesion Promoting Primer of Polyurea waterproofing membrane:

It is 1-K aromatic moisture curing PU adhesive, it can be used as a waterproof primer on polyurea surface. It has excellent bonding strength, weather resistance, hydrolysis resistance after fully curing. (about 3 days.)

Property	Result
Component	PU pre-polymer
Viscosity	700-900 cps/25°C
Solid Content	100%
Shelf Time	6 months

- F. “DP 3000 SBS -SA” sheet applied self-adhering SBS waterproofing membrane (min. 1.5mm thick) for area as specified in waterproofing schedule and demarcation drawings including but not limited to Planter / Landscape.

Other products submitted by the contractor may be considered for incorporation subject to compliance with the performance and properties, and approval by the Architect. The proprietary sheet applied self-adhering SBS waterproofing membrane shall meet the following minimum performance:

Properties	Test Methods	Results
Protection of the top side	high-performance polymer film	
Protection of the bottom side	self-adhesive binder / anti-adhesion film	
Straightness	EN 1848-1	≤ 10 mm / 5 m
Mass per unit area, kg/m ²	EN 1849-1	1.5±0.15
Thickness, mm	EN 1849-1	1.5±0.10
Tensile properties: maximum tensile	ASTM D5147	400±100 / 300±100

force L/T, N/50mm		
Tensile properties: elongation L / T, %	ASTM D5147	≥ 800 / ≥ 800
Crack bridging, mm	-	≥ 0.6
Determination of shear resistance of joints, kN/m	-	>110
Softening point, °C	ASTM D36	≥ +100
Flexibility at low temperature, °C	EN 1109-1	≤ -15
Flow resistance at elevated temperature, °C	EN 1110*	≥ +85
Visible defects	EN 1850-1	Pass
External fire performance	EN 13501-5	Froof
Watertightness at 60 kPa	EN 1928	Pass
Peel resistance of joints, KN/m	-	>3.3
Water vapour transmission properties	EN 1931	μ=90000
Dangerous substances	Does not contain dangerous substances	

- G. “DP 3000 SBS-4” sheet applied heavy duty, high strength, non-woven polyester mat impregnated and coated with a tough SBS modified bitumen waterproofing membrane (4mm thick) for area as specified in waterproofing schedule and demarcation drawings including but not limited to Flat Roof / Podium Deck / Planter / Landscape.

Other products submitted by the contractor may be considered for incorporation subject to compliance with the performance and properties, and approval by the Architect. The proprietary sheet applied SBS modified bitumen waterproofing membrane shall meet the following minimum performance:

TECHNICAL DATA	
Total thickness	4.0mm (nominal)
Straightness, mm	≤20 (EN 1848-1)
Mass per unit area, kg / m ²	5.1 ± 0.3 (EN 1849-1)
Cold flexibility, °C	≤ -25 (EN 1110 & ASTM 5147)
Flow resistance elevated temperature, °C /2h – upper face and lower face, °C	≥ 100 (EN1110 & ASTM D5147)

Softening point, °C	≥ 110 (ASTM D36)
Elongation, L/T, %	≥ 20 (ASTM D5147)
Tensile strength, L/T, N / 50mm	800 / 800 ± 200 (ASTM D5147)
Water absorption, %	≤1.0 (GOST 2678-94)
Tear strength, L/T, N	500 / 500 ± 150 (ASTM D4073)
Shear resistance of joints, L/T, a. N / 50mm b. kN / m2	a. ≥ 550 / 550 b. ≥ 110 (EN 12317-1)
Peel resistance of joints, max / average, kN / m	≥3.5 / 3.3 (EN12316-1)
Watertightness / 24 hours a pressure of 0.2MPa, kPa	Waterproof (GOST 2678-94)
Crack Bridging	≥ 1 mm
Visible defects	Pass (EN 1850-1)
Water vapor transmission properties	μ=20000 (EN 1931)
Dangerous substances	Does not contain dangerous substances
Properties after artificial ageing / EN 1296. 12 weeks at +70°C	
Flow resistance at elevated temperature °C / 2 h – upper face and lower face, °C	≥ 80 (EN 1110)
Cold flexibility, °C / Ø 30 mm – upper face and lower face, °C	≤ - 15 (EN 1109-1)

Remarks: Footnotes: L/T – Longitudinal / Transverse

- H. “DP Geotextile” the geotextile as protection layer or separation layer (0.8mm thick) for area as specified in waterproofing schedule and demarcation drawings including but not limited to Flat Roof / Podium Deck / Planter / Landscape.

Other products submitted by the contractor may be considered for incorporation subject to compliance with the performance and properties, and approval by the Architect. The proprietary protection layer or separation layer shall meet the following minimum performance:

items	Unit	Standard request	Note
Weight variation	%	-8	
Thickness	mm	0.8	
Width variation	%	-0.5	
Breaking strength	kN/m	≥2.5	CD/MD
Breaking elongation	%	25-100	CD/MD
CBR mullen burst strength	KN	≥0.3	
Sieve size O90(O95)	mm	0.07-0.2	
Tear strength	KN	≥0.08	

- I. “DP Brushcrete” polymer modified cementitious waterproofing coating in min. 2mm thick for water tank and in min. 1mm thick for all area as specified in water schedule and demarcation drawings including but not limited to water tanks, internal wet area such as lavatory, pantry, shower room, plant room, pump room, refuse storage room and external wall with rendering finishes and sunken slab for installation of turntable and road blockers etc.

Other products submitted by the contractor may be considered for incorporation subject to compliance with the performance and properties, and approval by the Architect. The proprietary polymer modified cementitious waterproofing coating shall meet the following minimum performance:

Typical Property	Result
Material Base	Polymer modified cementitious
Component	Two-part (liquid + powder)
Density (mixed)	1.90 g/ml
Crack bridging test method by ASTM C836-06	2.188mm
Pull off strength to concrete test method by BS EN 1348	1.04 MPa
Pull off strength by compatiability of SBR modified c/s render on the waterproofing membrane on concrete panel test method by BS EN 1348	0.90 MPa
Pull off strength by compatiability of c/s render on the waterproofing membrane on concrete panel test method by BS EN 1348	0.80 MPa
Adhesion strength to Yau Lee Wah panel wall by HKHA Specification Library Method WAT 6.M010.A	0.693 MPa

Clause 2(i)	
Compatibility with tile adhesive test method by BS 5980:1980 (AMD5861) : Appendix D with modification	3355N average (14 days under normal curing condition)
Compatibility with tile adhesive test method by BS 5980:1980 (AMD5861) : Appendix D with modification	3486N average (7 days under normal curing condition and followed by water immersion for 7 days)
Elongation test method by ASTM D412	567% average
Toxicity	Non toxic and Passed JC1066-2008
VOC	None
Temperature for application	+5 degree C or above
Water Pressure test method by DIN1048: Part 5 : 1991, 72 hours	5 bars (50m height) and 72 hours No sign of water penetration

- J. “BASF MasterProtect 300” (formerly known as “BASF Masterseal 300HS”) cold liquid applied anti-carbonation and anti-chloride elastomeric waterproofing membrane without bitumen content in min. 0.5mm thick for external wall behind cladding.

Other products submitted by the contractor may be considered for incorporation subject to compliance with the performance and properties, and approval by the Architect. The proprietary anti-carbonation and anti-chloride elastomeric waterproofing membrane without bitumen content for external wall behind cladding shall meet the following minimum performance:

- i. Designed for the protection of concrete structures against carbonation and chloride ingress.
- ii. The product is also suitable as a seamless and elastomeric waterproofing coating for timber, asbestos / fiber cement and zinc sheets, asphalt, build-up felt and tiles.
- iii. Easily applied by roller, brush or airless spray.
- iv. Flexible – capable of bridging cracks.
- v. Protective – barrier against salts and atmospheric gases.
- vi. High-build – masking imperfections in substrates.
- vii. Waterproof – protects concrete from waterborne salts.
- viii. U.V. stable – maintains its appearance.
- ix. Without bitumen content – no risk of bitumen leakage and staining.
- x. Typical Properties:

TYPICAL PROPERTIES	TEST METHOD	RESULT
Relative density	---	1.35 at 25°C
Moisture vapour transmission	BS 3177	1.98 g/m ²
Chloride ion diffusivity	Taywood	4.98 x 10 ⁻¹⁰ cm ² /s
Reduction in chloride ion ingress @28 days	---	At least 97%

Carbon dioxide diffusion	Taylor Woodrow Lab	R value at 325 microns greater than 161m
Application temperature	---	5°C to 50°C
Chemical resistance	---	Resistant to spillage of gasoline, diesel, sewage, weak acids and alkalis
Elongation	ASTM D412:2006	≥ 434%
Tensile strength	ASTM D412:2006	≥ 3.5 N/mm ²
Tear strength	ASTM D624	7.3 kN/m average
Accelerated weathering	BS 3900 : Part F3	Passed
UV Exposure	ASTM G154:2000a Fluorescent UVA340nm for 500 hours and 1,000 hours	No disintegration, blistering, peeling, cracking or crazing
Natural weathering for 2 years	---	No cracking, crazing, flaking, lifting, formation of blister, erosion or yellowing
Fire Exposure Test	BS 476:Part 3	Class 'EXT.S.AA'
Bond strength to concrete with Masterkure 128 primer	QCC/LP26-02	> 2.7 N/mm ²
Crack bridging	BRE	5mm
Hardness (Shore A)	ASTM D2240:2005	55
Anti-microbial activity	Thor Chemicals (UK)	Passed a. Wet State Bacterial Examination b. Wet State Fungal Test Procedure c. Film Fungal Protection Test
Potable water use	EEC Directive 80/778/EEC	Toxic metal test below the Maximum Acceptable Concentration
	Water Research Council (WRC)	Approved
Colours	---	Light grey and white
Dynamic Impact Test	No any sign of damage of failure in adhesion of the pedestal	A soft body comprising a canvas/ leather bag containing dry sand with different weight (1 to 5kg) was arranged to swing through an arc of 90 degrees of a radius of 1metre to strike the head of

		selected pedestal
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- K. "PCI Seccoral 2K Rapid" trowel-liquid applied flexible and crack-bridging anti-efflorescence barrier with waterproofing function in 1.5mm thick for external wall with tiling finishes.

Other products submitted by the contractor may be considered for incorporation subject to compliance with the performance and properties, and approval by the Architect. The proprietary trowel-liquid applied flexible and crack-bridging anti-efflorescence barrier with waterproofing function for external wall with tiling finishes shall meet the following minimum performance:

- i. The system is compatible with all other cementitious typed contacting surfaces and tile adhesive materials and it is verified by the compatibility test at item ii as below;
- ii. Standard grade of waterproofing system shall have the following typical properties when cured for 28 days.

Property	Test Method	Typical Value
Pull-off strength with adhesive after storage in 23°C and 50% R.H. for 28 days	ZDB code of practice	$\geq 0.9 \text{ N/mm}^2$
Pull-off strength with adhesive after storage in water for 21 days;	ZDB code of Practice	$\geq 0.6 \text{ N/mm}^2$
Pull-off strength with adhesive strength after warm storage (70°C) for 14 days;	ZDB code of practice	$\geq 1.1 \text{ N/mm}^2$
Pull-off after Storage in chloride water for 7 days;	ZDB code of practice	$\geq 0.9 \text{ N/mm}^2$
Pull Off Strength after storage in lime water $[\text{Ca}(\text{OH})_2]$ for 7 days	ZDB code of practice	$\geq 0.7 \text{ N/mm}^2$
Crack bridging properties	DIN EN1062-7 (Method)	A4 (0°C at 2mm)
Waterproofing properties	ZDB code of practice	1.5 bar no water penetration (15m water pressure)
Bonded waterproofing directly under tiles for indoor and outdoor use, edition 2005	ZDB code of Practice	Class I, II and III
Waterproofing + tile adhesive + epoxy tile grout (PCI Seccoral + PCI Nanolight + PCI Durapox NT)	KSW recommendations (Hygiene Institute Gelsen Kirchen)	Comply
Waterproofing of buildings and structural elements under ceramic tiles using PCI deformable tile adhesive	Technical University of Munich according to Construction Products list A, Part 2, no. 1.10 by DIBt	Class A1, A2 and B
Shore A	ASTM D2240	≥ 88 average
Water Vapour Permeance	ASTM E96 (Method B)	≤ 0.24 perms
Resistance to fire	DIN 4102-1	ClassB2
Resistance against turbine oil 1. 40°C warm oil at 1.5 bar pressure 2. 7 weeks of exposure to oil at 1.5 bar pressure	BPG	No swelling or softening and no penetration observed

Resistance against insulating oil	HKHA specification WAT6.M010.4	No leakage
Expected Service Life	European Technical Approval ETA-07/0177	At least 25 years
Drinking Water application	DVGW W270 and W347	Pass
Control of Emission in Products	GEV-EMICODE	Class ECI Plus
Liquid-applied water impermeable product for installation underneath ceramic tiling	EN14891:2012 + AC:2012	Class System 3
Liquid applied watertight covering kit for wet rooms floors and wall	ETAG022 Part 1 : 2007	Class System 2 + System 3 (Reaction to fire)

- The system has a crack-bridging capability for bridging cracks up to 1mm even if cracks are subsequently occurred in the substrate after the waterproofing system has been cured;
- The system is ready to accept tiles, fixed with the appropriate "PCI" Tilefast adhesive, after 1 day;
- The system can be applied in a cold liquid form ;
- The system must incorporate PCI Pecitape special reinforcement materials at the weakest areas such as the wall to floor junctions, any pipe extrusion and outlets, so as to discourage the water leakage at these areas even under the stress of substrate and thermal movement ;
- PCI Pecitape special reinforcement and flexible waterproofing tape must have the following typical properties:

Parameter	Standard	Test Result
Material		TPE coated Polypropylene Fleece
Weight	EN 1849-2	315 g/m ² - 400 g/ m ²
Thickness	EN 1849-2	> 0.5 mm
Width	EN 1849-2	117 mm – 123 mm
Straightness	EN 1849-2	< 50 mm
Flatness	EN 1849-2	< 15 mm
Tensile strength lengthwise	EN ISO 527-3	> 80 N / 15 mm
Tensile strength crosswise	EN ISO 527-3	> 15 N / 15 mm
Tensile stress at break lengthwise	EN ISO 527-3	> 40%
Tensile stress at break crosswise	EN ISO 527-3	> 200%
Resistance to tearing lengthwise	EN 12310-2	> 15 N

Resistance to tearing crosswise	EN 12310-2	> 120 N
Water tightness	EN 1928	> 1.5 bar

- PCI Pecitape special reinforcement and flexible waterproofing tape must be compatible with proposed waterproofing system with the minimum pull off test of 0.5N/ mm² according to EN24642;
- The system is proof against water penetration at up to 15m hydrostatic head ;
- All components of the waterproofing system are non-toxic and do not cause any hazard or injury to operatives or to others in the vicinity of the Site;
- Where waterproofing system is found to be defective or damaged, even if the damage is due to cracks on substrate subsequently developed after application of water proofing system, the system must allow for repairing works to be carried out locally within an area not more than 300mm measured from the defective or damaged area. And the water tightness performance of the whole waterproofing system after carrying out of repairing work shall be unaffected.

L. "MBS Masterseal 550" waterproofing membrane

Other products submitted by the contractor may be considered for incorporation subject to compliance with the performance and properties, and approval by the Architect.

Density:	1826kg/m ³
Toxicity:	Non toxic - Suitable for use in contact with potable water
Initial surface absorption:	>95% reduction against control
Chloride ion diffusivity:	Not measurable after 24 months of testing
Moisture vapour transmission (equivalent air layer thickness):	SD ≤ 1.5m (839µm dft)

- All components of the waterproofing system are non-toxic and do not cause any hazard or injury to operatives or to others in the vicinity of the Site;
- The system is available in 20kg double pack
- The system is available in three standard colours : Light Grey, White and Dark Grey
- The system can be applied to 24 hour-old concrete thereby giving immediate protection.
- The system is excellent adhesion. Bonds to porous and non- porous surfaces.
- The system is suitable for light pedestrian traffic
- The system is breathable - whilst repelling water, allows substrate to breathe
- The system is high resistance to chloride ion diffusion
- The system is composed of specially selected cements, silica sand and reactive fillers supplied in powder form together with a liquid component of blended acrylic copolymers and wetting agents
- The system follows the standard of WRAS - "Suitable for use in contact with potable water" - BS 6920 : 2000
- The system provides an effective barrier to waterborne salts such as chlorides and sulphates.
- The system is outstanding wear and weather resistance and good resistance to gasoline, diesel oil, sodium hydroxide, calcium chloride, de-icing salts
- The system coated surfaces exhibit good resistance to mild acids.
- The system is an extremely effective barrier to atmospheric acidic gases which cause

carbonation in concrete structures.

- The system is at an applied rate of 1.8 kg/m² gives an equivalent air layer thickness for carbon dioxide diffusion (R) of 92 metres. The accepted minimum value for R is 50m. Testing to confirm this was carried out independently by Taywood Engineering 2005
- The system is 1.826 kg / m² at 1 mm thickness

PART 3 EXECUTION

3.01 PREPARATION

- A. Concrete surfaces shall be a floated finish, free from cavities and projecting nibs, shuttering faces shall be of an equivalent standard, damaged concrete shall be made good with waterproof cement sand mortar.
- B. Smooth brick and block surfaces shall be sound and course joints flush pointed. Rough brick or open texture blocks shall be made smooth by application of a waterproof sand cement render.
- C. All surfaces shall be frost free, dry and free from dust, dirt, oil, curing compounds and other harmful foreign matter. The Contractor shall submit a Letter of Confirmation that all water retaining structures and tanks, concrete walls and slabs are free of curing compounds, and material compatibility testing have been conducted with satisfactory results, prior to the waterproofing installation.
- D. Liquid Applied Waterproof Membrane:
 - 1. Remove oil, grease, wax, paint, adhesive, sealer, curing compounds and other foreign substances from concrete surfaces and slabs with a mechanical scarifier. A strong detergent solution such as trisodium phosphate or a solution of lye and hot water may also be used. The solution shall be mopped on the surface and allowed to stand 10 or 15 minutes, or until foreign substances are loosened. The solution shall then be squeezed or mopped off and the treatment repeated. When all of the material has been removed from the surface and floor, the surface shall be flushed thoroughly with clean water to remove any remaining cleaning solution.
 - 2. Repair shrinkage or non-structural cracks with proprietary crack suppression kit, consisting of a liquid rubber and 150 mm reinforcing fabric tapes or other approved methods recommended by the manufacturer of liquid applied waterproof membrane.
 - a. Apply liberal coat of liquid with paint brush or roller over crack, slightly wider than width of reinforcing tape, roll out enough fabric to cover crack. Cut and imbed while liquid is wet. After laying fabric into liquid, firmly brush or roll a top coat of liquid over tape. When surface is dry to the touch, it is ready for applicator of the finished surface.
 - b. For large cracks to 3 mm wide, 2 layers of fabric shall be used.
- E. Roofing Works
 - 1. Verify that all substrate and deck are clean and smooth, free of depressions, waves, or projections, properly sloped to drains, valleys, and eaves. Remove weak laitance, loose particles, debris, dirt and dust by vacuum equipment.
 - 2. Verify all roof openings, sleeper walls, plinths, curbs, pipes, sleeves, ducts, vents, façade maintenance unit guide rail fixing dowels, and other penetrations, through roof and deck are solidly set, all cant strips, nailing strips, and reglets are in place. Verify deck is supported and tightly secured.
 - 3. Verify structural roof deck surfaces are dry. Confirm dry deck by moisture meter with 12% moisture maximum, or as recommended by the Manufacturer.
 - 4. Beginning of installation means acceptance of substrate, and Project conditions. Pre-Installation Conference has been held with agreements reached, and roof mock-up has been approved.

5. Building surfaces and equipments are protected against damage from all roofing work.
6. Provide temporary, removable safety rails around all perimeter roof edges not protected with guardrails or 1100 mm high parapets. Workmen's safety harness, helmet, and other protective gears shall be worn by all roof installer to meet all relevant BS and ISO construction safety standards.
7. Thoroughly inspect and test operate all roofing torch equipment and fire extinguishers prior to start of any roof work to ensure fire safety.
8. Prior to the installation of the roofing membrane, the contractor must check if the roof substate complies with the Manufacturer design instructions. Any defects in the substrate need to be corrected and the final surface has to be prepared to meet the requirements previously outlines. It is important that the substrate is free of any sharp objects and / or products that may damage the membrane. Wipe the substrate and install a protection fleece when required.
9. Allow ample material for splicing with the overlap of adjoining sheets determined by the type of seam and tie-ins. The tie-in material will be determined by the applicable detail. Provide an extra 150mm of membrane at roof edges and wall terminations to facilitate final positioning.
10. Temporary ballasting during installation may be required to keep the membrane in place until it is secured to the substrate. Suggested temporary ballasting includes sand bags and other non-abrasive materials such as rubber tires, etc. Never leave the project without temporary ballasting the loose laid sheets.

3.02 INSTALLATION/APPLICATION

A. Liquid Applied Waterproof Membrane:

1. Install waterproof membrane once surface of crack pre-treatment has dried to the touch. Use a wide paint roller to cover substrate with liberal coat of waterproofing. Cover an area slightly wider than the reinforcing fabric width. Imbed fabric while liquid is still wet. Apply additional liberal coat of waterproofing liquid to cover and saturate fabric. Lap fabric 50 mm at seams. Allow top coat of waterproofing has dried to the touch, install finishes. Waterproofing shall be applied 200 mm up against all floor-to wall and floor-to-curb/obstructions, unless noted otherwise.
2. Install waterproof membrane to the required dry film thickness in strict accordance with Manufacturer's installation instruction.

B. Installation of Protection Boards:

1. Pour the mixed mastic compound on to the surface and spread with a squeegee at a rate of 10-12 sq metres/unit depending on surface texture and temperature. Protection boards shall be laid while the compound is still wet and laid progressively to minimise applicator trafficking until the compound has cured. The boards must be close butt jointed to ensure continuous protection, but where gaps occur these shall be filled with mastic compound prior to applying the reinforcing tape to ensure watertightness.
2. Where protection boards abut parapets, pipe bays or abutments they shall be pre-measured, and accurately cut to size by incising with a sharp knife and breaking on edge.
3. The joints between the protection boards shall be coated with primer, applied in 100 mm wide strips with a brush or roller and allowed to dry before applying the self-adhesive tape centrally over the joint.

4. The reinforcing tape must be firmly roller with a lap roller along its length and at junctions to ensure a watertight joint. Seal the exposed edge of the boards at the end of each working period to prevent the ingress of moisture overnight, by tooling the compound against the exposed edge of the boards. Ensure that the boards are bonded to the substrate by firmly rolling or light trafficking before back filling or installing finishes above.
5. Base and wearing courses shall be applied as soon as possible after a minimum of four hours. Bituminous material with a temperature greater than 145°C shall not be laid or deposited on deck waterproofing systems, unless precautions against heat damage have been approved by the Waterproofing Manufacturer's Representative.

C. Installation of Tank Waterproofing Protective Coating

1. Surface Preparation

- a. All surfaces to be coated must be structurally sound and dry, free from loose material and grease or oil.
- b. Concrete surfaces: New concrete shall be allowed to cure for at least 28 days prior to priming. Prepare concrete by blast cleaning or water jetting, ensuring that all blow-holes are opened up. These shall then be filled with patching filler. Oil and grease contaminated surfaces shall be treated with a proprietary degreasing agent followed by washing with clean water or steam cleaning.
- c. Metal Surfaces: Any rust, mill scale or other impurities shall be removed from the substrate by grit blasting to bright metal.

2. Priming

- d. Concrete surfaces: All prepared concrete surfaces shall be primed and thoroughly dampened with water.
- e. Metal Surfaces: All metal surfaces shall be primed immediately after preparation.
- f. Mixing: In cold weather materials shall be stored at 15°C for at least 24 hours before use. The contents of both cans shall be stirred thoroughly to disperse any settlement. Mix both components till a homogeneous mixture is obtained. It is important that both components are intermixed thoroughly and that no traces of the components remain unmixed. A uniform grey colour shall be obtained.

3. Coating

- g. Apply tank coating where applicable to the primed surface immediately after mixing. Use a nylon brush and finally smooth out using a steel trowel. A continuous coating of uniform thickness shall be obtained. Apply subsequent coats perpendicular to the direction of the application of the previous coat after the previous coat has dried for min. 4 hours.

4. Use of Glass Fibre Reinforcement

- h. Tank coating, where applicable, shall be used in conjunction with glass fibre cloth to increase thickness, or where necessary, bridge fine cracks in the substrata. The cloth shall be laid directly onto the first coat whilst it is still wet and shall be pressed in and smoothed out with a stiff nylon brush or split washer roller. Second and subsequent coats shall be applied, allowing 16 hours (but no more than 24 hours at 20°C) between each coat.
- i. Suitable cloth is open weave 110g/m² glass cloth.

5. All other issues can be carried out strictly in accordance with Manufacturer's written instructions, to be submitted to the Architect.

D. Vapour Barrier Installation (if required):

1. Install vapour barrier within the Roof System Assembly if recommended by the Manufacturer, and at the proper location within the assembly confirmed by him.
2. Ensure that all vapour barrier are undamaged and imperforated. All joint laps shall be 75 mm minimum and sealed with reinforced tape. Longitudinal and transverse lap joints shall be staggered. Maintain vapour tightness at all lap joints and throughout vapour barrier. Conform with BS6229 and BS5250.
3. At parapets, up stands and abutments, vapour barrier shall be turned back over the insulation board and stuck down, or carried up and dressed with roof screed or render.
4. Do not place screed above vapour barrier due to risk of water absorption during construction.

3.03 CLEANING

- A. At completion of the roof system assembly work, remove all temporary protection and barricades from all decks and roofs, and the Project Site.
- B. Clean all items and adjacent surfaces roof work. Examine all roof work and repair all damage. In the event damage is irreparable, remove and replace such items at no additional cost. Do all required touch-up to marred or abraded surfaces. Leave entire Work in condition acceptable to the Employer.
- C. The Contractor shall at all times keep the Project roof and deck area free of excessive waste, materials and rubbish. Prior to completion of all roof work, the Contractor shall remove all such waste, equipment, other property and tools from the Project Site. Legally and safely handle, transport, and dispose of waste and debris.
- D. Remove all debris after completion of the Work of this Section. Store left over materials as directed by the Architect.
- E. Clean adjacent surfaces and materials soiled by any work of this Section.

3.04 PROTECTION

- A. Ensure that all waterproofing are protected from traffic, weather and damage during construction. Repair or replace all damaged waterproofing before covering with other finish materials.
- B. Protect all insulation from damage from wetting, loading or abuse until they are under protective covering material or finish.
- C. Protect adjacent surfaces and materials from damage and soiling by the work of this Section.
- D. The Contractor shall continually protect Roof System Assembly from all damage and traffic during and after roof installation work. Install and provide temporary traffic pads on the roof for work of other trades during Project Construction.

3.05 MAINTENANCE

A. ROOF MAINTENANCE & MANUAL

1. Maintenance Manual

- a. Upon completion of the roof system assembly work, the Contractor in conjunction with the Roofing Manufacturer shall prepare a roof maintenance manual, identified with project name, location and date. Identify each type of roofing material, and accessories used, methods applied and surface to which system was applied, and provide sketches where necessary. Recommendations for periodic inspections, care and maintenance shall also be included in the Roof Maintenance Manual. Identify common cause of damage with instructions for temporary patching until permanent repair can be made.
- b. One copy each of the Roof Maintenance Manual (with periodic inspection check lists) shall be given to the Employer, the Project Building Facility Manager, the Roof Manufacturer and the Architect. The Roof Maintenance Manual shall be translated into Chinese and English. All cost associated with the preparation and distribution of all Roof Maintenance Manuals shall be included in the Contract Sum.

2. Maintenance & Emergency Calls

- a. During two years following, the date of Practical Completion of the Project, the Contractor shall agree to respond to all Employer's reported roof leaks or related other problems within 48 hours to execute repairs.
- b. During two years following the date of Practical Completion of the Project, the Contractor shall thoroughly inspect and provide routine maintenance work every 4 months on all roofs and decks of the Project. A roof and flashing inspection shall be conducted within 24 hours after the first occurrence of a major typhoon, monsoon, or rainstorm at the Project Site. Report and rectify all damages as required to continually maintain a weather tight (both water tightness and air tightness) roof system assembly and sheet metal flashing for every roof and deck on the Project.

3. Roofing System Notices

- a. A notice describing the roofing system assembly installed on the Project with maximum allowable deadloads, live loads and other structural limitation of all roofs and decks shall be posted conspicuously at all roof and deck access locations. The notice shall state that no new deck or roof equipment and penetration, or any modification, may be made without prior review by a Structural Engineer, an Architect and the Contractor and Roof System Manufacturer to ensure continued warrantability and weather tightness of Project roof and deck.

END OF SECTION